Department of Electronics Engineering

Walchand Institute of Technology,Solapur

A Data Analytics Report

(CouseCode:23ECU6CC2T)

ON

DATATHON

Submitted by

1. Aditya Jain	**Roll No. 20**	**PRN No. 2303111026**
2. Om Jamage	**Roll No. 23**	**PRN No. 2303111048**
3. Dhruv Kanhed	**Roll No. 29**	**PRN No. 2303111062**

Department of Electronics Engineering,
Walchand Institute of Technology, Solapur

(*An Autonomous Institute*)

Ashok Chowk, Solapur 413006

YEAR- 2025-26

Department of Electronics Engineering

Walchand Institute of Technology,Solapur

CERTIFICATE

Thisis to certifythat theMini Project entitled Datathon is a bonafide work carried out in partial fulfilment of the requirements of the Data Analytics course during the academic year 2025-26.

Submitted by

1. Aditya Jain	**Roll No. 20**	**PRN No. 2303111026**
2. Om Jamage	**Roll No. 23**	**PRN No. 2303111048**
3. Dhruv Kanhed	**Roll No. 29**	**PRN No. 2303111062**

- **Class: T.Y B. Tech (Electronics and Computer Engineering)- Semester – VI**
- **Course Name: Data Analytics Course Code: *23ECU6CC2T***

The report has been approved as it satisfactorilyfulfils the academic requirements prescribed for the Mini Project course.

……

Mr.R.P.Nagarkar

In-charge faculty

ABSTRACT

Maintaining a healthy lifestyle has become increasingly important, yet many individuals struggle to create personalized fitness plans suited to their body and goals. This project presents FitAI – Fitness Recommendation System, a machine learning-based application that generates customized fitness insights including BMI calculation, weight classification, and ideal weight recommendations.

The system utilizes Decision Tree Classifier and Linear Regression models trained on a dataset of 3,864 fitness records. The application is implemented using a Streamlit-based interactive UI, allowing users to input personal details such as age, gender, weight, height, and exercise duration.

The system provides real-time outputs such as BMI value, weight category (underweight, normal, overweight), and ideal weight range. This project demonstrates how machine learning can be applied to healthcare and fitness domains to provide intelligent, user-friendly solutions.

1. INTRODUCTION AND PROBLEM DEFINITION

1.1 Background

Intoday's fast-paced lifestyle, people often lack access to personalized fitness guidance. Generic workout plans fail to address individual needs based on body metrics, leading to ineffective results.

Machine learning enables the creation of intelligent systems that can analyze user data and provide tailored fitness recommendations.

1.2 Problem Statement

Mostindividuals do not know:

- Their exact BMI status
- Whether they are underweight, normal, or overweight
- Their ideal body weight
- How their lifestyle affects fitness

There is a need for an automated system that can analyze user inputs and generate meaningful fitness insights instantly.

1.3 Objectives

- To calculate BMI using user input data
- To classify users into fitness categories using ML models
- To predict ideal weight using regression techniques
- To build an interactive UI using Streamlit
- To provide real-time personalized fitness recommendations

2. SYSTEM ARCHITECTURE AND METHODOLOGY

2.1 Module Description

Thesystem consists ofthree main modules:

1. **Machine Learning Module**

- Data preprocessing
- Model training (Decision Tree & Linear Regression) Model
- saving using joblib

2. Prediction Module

- Loads trained models
- Processes user inputs
- Generates predictions

3. Frontend UI Module

- Built using Streamlit
- Takes user inputs via sliders and fields
- Displays BMI, category, and recommendations

2.2 Working Principle

- User enters details:
- Age
- Gender
- Weight
- Height
- Exercise duration
- System calculates BMI
- Input data is passed to ML models
- Output generated:
- BMI value
- Weight category
- Ideal weight range
- Results displayed instantly on UI

2.3 Algorithm Methodology

- Decision TreeClassifier
- Used to classify users into categories:
- Underweight
- Normal
- Overweight
- Linear Regression
- Used to predict:
- Ideal weight based on height and other factors

3. DATA PROCESSING AND ENGINEERING

3.1 Dataset Description

- Dataset contains3,864records
- Features include:
- Age
- Gender
- Height
- Weight
- Exercise duration
- BMI

3.2 Preprocessing and Feature Engineering

- Handling missing values
- Encoding categorical variables (Gender)
- Feature scaling (if required)
- Splitting dataset:
- 80% training
- 20% testing

3.3 Design Considerations

- Removed irrelevantdata
- Focused on key fitness parameters
- Ensured real-time performance
- Designed simple and user-friendly UI

4. SOFTWARE DESIGN AND IMPLEMENTATION

4.1 Tools Used

- Python 3.x
- Scikit-Learn, Pandas, NumPy
- FastAPI
- Streamlit
- Joblib

4.2 Code Explanation and Integration

Model Training

- Train Decision Tree & Linear Regression
- Save models using .pkl files

Frontend (Streamlit)

- Sidebar inputs
- Dynamic UI updates
- Displays BMI & recommendations

5. TESTING, RESULTS AND DISCUSSION

5.1 Testing Procedure

- Testedwith different user inputs
- Checked edge cases (low/high values)
- Verified UI responsiveness

5.2Results and Analysis

- BMI calculatedaccurately
- Correct classification (Underweight/Normal/Overweight)
- Ideal weight range displayed
- Real-time response achieved

6. CONCLUSION AND FUTURE SCOPE

6.1 Summary

TheFitAIsystem successfully demonstrates how machine learning can be used to provide personalized fitness recommendations. It simplifies complex calculations and delivers meaningful insights in an easy-to-use interface.

6.2 Future Scope

- Add diet recommendation system
- Integrate deep learning models
- Connect with wearable devices (smartwatch)
- Deploy as a mobile app or web platform
- Add AI chatbot for fitness guidance